Search with Job-Provided Referral Opportunities

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Abstract

This paper explores the impact of job-provided networking and referral opportunities on labor market outcomes. Using data from the Survey of Consumer Expectations, I estimate a continuous-time job search model with bargaining in which workers encounter jobs through either a referral or official channel, and jobs differ in their ability to generate referrals during on-the-job search. The model is estimated separately for a less-than-college and a college degree sample. I find that job-provided networking increases wages and employment, with effects that are significantly larger for workers without a college degree. Across both education samples, workers in low match-productivity jobs benefit the most from being in a job with network opportunities, because they increase exposure to higher-paying jobs.

1 Introduction

Referrals and interpersonal networks play an important role in labor market outcomes and are well documented by the surveys of Ioannides and Loury (2004) and Topa (2011). A key finding of this literature is that referral-based search channels shape not only job-finding rates but also the progression of workers over the course of their careers. By transmitting information about vacancies and by providing some form of additional screening, networks affect match quality, job tenure, and the sequence of jobs workers take (Calvó-Armengol and Jackson, 2004; Montgomery, 1991). Importantly, access to referral opportunities is uneven: some workers are better connected, and some jobs generate more exposure to networks than others. This heterogeneity implies that referrals can influence long-run wage growth and job mobility to better jobs, rather than merely improving match quality and the wage at referred jobs (Galenianos, 2014).

In the empirical literature, Brown et al. (2016) use data from a large U.S. firm to show that referred candidates have higher hiring probabilities, initially higher wages, and lower turnover. Schmutte (2015) uses matched employer–employee data to show that referral networks affect both job assignment and earnings. Baranowska-Rataj et al. (2023) similarly emphasizes that low-skilled workers with high-skilled co-worker networks are more likely to move to better-paid jobs as a result of the network and peer effects.

In looking at the heterogeneous usage and impact of referrals, Loury (2006) finds that referred

workers tend to have longer tenures, and attributes these longer tenures to either better matches via the referral channel, or rather, that workers who use referrals tend to have fewer outside options. Using a model with heterogeneous network structures, Arbex et al. (2019) shows that better-connected workers earn higher wages, climb the job ladder more rapidly, and rely more heavily on referrals at higher rungs of that ladder. Lester et al. (2021) further explores heterogeneity along both worker and referral dimensions, splitting the referrals channel into referrals from friends and family and referrals from business contacts. This distinction is important in their calibrated model, as they find referrals from friends and family primarily facilitate employment for workers who struggle to obtain offers through formal channels, while business referrals disproportionately raise wages for workers with already strong labor market opportunities.

This paper contributes to this literature by modeling the impact of work-provided networking opportunities on wages and job-to-job transitions. Using data from the New York Fed's Survey of Consumer Expectations Job Search Supplement (Armantier et al., 2016), which records referral offers, referral contacts, and referral-based hires for both employed and unemployed individuals, I estimate a model where jobs differ in their ability to generate referral opportunities. This is done by having workers in "network jobs" and "non-network jobs" have different arrival rates for job offers from a referral channel. The objective of this model is to explore how specific jobs provide more access to networking options that can lead to subsequent jobs, and to understand which groups of workers benefit from these jobs the most, and what happens to their wages both at these jobs and subsequent wage growth. The model heavily builds on the structural framework from Lester et al. (2021), but omits worker ability and the distinction between referral types.

2 Model

2.1 Environment

Time is continuous. Workers are risk–neutral and discount at rate $r > 0$. Each job is characterized by a type $n \in \{0, 1\}$ and a match quality $x \in [x_{\min}, x_{\max}]$. Job types are chosen exogenously, where for any given job, its probability of being type n is π_n. Conditional on receiving an offer of type n, the match quality, x, is drawn from a type–specific beta distribution G_n on $[x_{\min}, x_{\max}]$, with density g_n. Output in a job with match quality x is linear and does not depend on job type,

$$f(x) \;=\; px + c, \tag{1}$$

Employed workers face an exogenous destruction hazard $\delta > 0$ that does not depend on job type.

2.2 Job Arrivals

In this model, workers received job offers through one of two channels. Either the worker is referred to a job by a friend, colleague, or business contact, or through a catch-all 'other' or official channel, which encompasses all other forms of job acquisition other than a direct referral. I denote these channels as R and O respectively. Unemployed and employed workers receive offers through the official channel at rates λ_O^U and λ_O^E. Job offers are generated via the referral channel at rates λ_R^U

and $\lambda_R^E(n)$, where for the employed worker, their offer rate is a function of their current job type. Thus, unemployed workers receive job offers at a total rate

$$\lambda^U = \lambda_O^U + \lambda_R^U, \tag{2}$$

where the arrival rate of offers for jobs of type n is $\pi_n \lambda^U$ with $\pi_0 + \pi_1 = 1$. Employed workers receive on–the–job offers at a rate

$$\lambda^E(n) = \lambda_O^E + \lambda_R^E(n), \tag{3}$$

again with type $n' \in \{0, 1\}$ of the incoming offer drawn according to probabilities $\pi_{n'}$, and match quality x' drawn from $G_{n'}$.

2.3 Value Functions and Wages

Let V^U denote the expected discounted flow value of an unemployed worker, and let $V^E(x, n, w)$ denote the discounted flow value of a worker in a job of type n with match-productivity x earning a wage w. Wages are determined through Nash bargaining, with the relevant outside option depending on the worker's employment status. For unemployed workers, the outside option is unemployment, while for employed workers, the outside option reflects competition between the current employer and a poaching firm. This wage-setting environment corresponds to a three-way bargaining problem involving the worker, the incumbent firm, and the poaching firm, as in standard on-the-job search models (Postel-Vinay and Robin, 2002)

Acceptance Rules:
An unemployed worker accepts a job offer of type n with match productivity x if,

$$V^E(x, n, f(x)) \geq V^U \tag{4}$$

This inequality defines a type-specific reservation match productivity, $x_n^\star$, satisfying

$$V^E(x_n^\star, n, f(x_n^\star)) = V^U \tag{5}$$

When an unemployed worker accepts an offer from a firm of type n with $x \geq x_n^\star$, they will earn a wage $w^u(x, n)$ that satisfies

$$V^E\Big(x, n, w^u(x, n)\Big) = V^U + \alpha(V^E(x, n, f(x)) - V^U) \tag{6}$$

Where $\alpha \in [0, 1]$ denotes the bargaining power of the worker, or simply, the share of the surplus value relative to unemployment that the worker receives. In the case of an employed worker currently in a job (x, n), they accept an outside offer, (x', n') if

$$V^E\Big(n', x', f(x')\Big) \geq V^E\Big(n, x, f(x)\Big) \tag{7}$$

Let $\mathcal{A}(x, n)$ denote the set of offers (x', n') that satisfy the acceptance condition. When an employed worker transitions to a new job, the wage $w'(n, x; n', x')$ depends on both the characteristics

of the current job (n, x) and those of the prospective job (n', x'). This reflects a three-way bargaining environment involving the worker, the incumbent firm, and the poaching firm. Before bargaining with the new firm, the incumbent firm competes by bidding the worker's wage up to the full marginal productivity, $f(x)$. Thus, the worker's outside option value equals the value of remaining employed at (n, x) while earning $f(x)$. Given this outside option, the wage, $w'(n, x; n', x')$ at the new job satisfies

$$V^E\Big(n', x', w'(n, x; n', x')\Big) = V^E(n, x, f(x)) + \alpha\Big[V^E(n', x', f(x')) - V^E(n, x, f(x))\Big]. \tag{8}$$

Value Functions:
For an unemployed worker, their value function is

$$rV^U = b + \lambda^U \sum_{n\in\{0,1\}} \pi_n \int_{x_n^\star}^{\overline{x}} \Big(V^E(x, n, w^u(x, n)) - V^U\Big)\, dG_n(x). \tag{9}$$

Where for a worker discounting at rate r, the value of unemployment is equal to the unemployed flow value, b, plus the option value of search. At rate λ^U, an unemployed worker receives job offers across job types $n \in \{0, 1\}$ and match qualities drawn from the corresponding distributions, and the value V^U reflects the average gain from offers that are optimally accepted.

$$\begin{aligned}(r + \delta + \Gamma(x, n))\, V^E(x, n, w) &= w + \delta V^U \\ &+\lambda^E(n) \sum_{n'\in\{0,1\}} \pi_{n'} \int_{\mathcal{A}(x,n)} \Big(V^E(x', n', w') - V^E(x, n, f(x)\Big)\, dG_{n'}(x')\end{aligned} \tag{10}$$

With $\Gamma(x, n) = \lambda^E(n) \sum_{n'} \pi_{n'} \int_{\mathcal{A}(x,n)} dG_{n'}(x')$

The effective discount rate for the employed worker is r plus the exogenous job destruction rate, δ, plus the arrival rate of on–the–job offers times the probability that an incoming offer (n', x') is accepted, aggregated across job types, represented by $\Gamma(x, n)$. The value of employment is equal to the wage w plus the discounted continuation value of job destruction, plus the surplus value of acceptable offers received while on-the-job times the arrival rate given the type of the current job.

3 Estimation

3.1 Data

To estimate the model, I use data from the Job Search Supplement (JSS) from the Survey of Consumer Expectations (SCE) (Armantier et al., 2016). The JSS was administered annually between 2013 and 2021 and includes a repeated cross-section of 10,311 U.S. adults' labor-force status, job search behavior, job offers, referral channels, and wages. The survey asks currently employed respondents to report the channel through which they learned about their current job, including separate categories for referrals from "a friend or relative," "a former co-worker, supervisor, or business associate," and "a current employee at the company." The survey also captures ongoing job search activity for both employed and unemployed individuals by asking how many offers they

have received in the past four weeks, and through which channels these offers were received. For the model, having all offers reported, not just the accepted offers as is often the case, means we can directly discipline the arrival rates of job offers by channel.

3.2 Assumptions

To simplify estimation, I fix the worker's bargaining weight, $\alpha = 0.5$, which provides the worker with half the surplus generated relative to their outside option. The discount rate r is set to 0.04. The largest difficulty with estimating the model from the JSS data is identifying job type. Since the networking capacity provided by the job is rather intangible, the type of the job must be determined by a variable, or a series of covariates, in the data. Ultimately, I chose to set a job type to $n = 1$ if the worker was themselves referred to that job by a business associate from the incumbent firm. The rationale being that for the worker to have encountered this business associate, the associate must be most likely in a job that actively involves networking with other firms and work. This is far from the ideal way to identify a network job, as a better way to identify these jobs is by some measurement of how client-facing the role is, whether or not the job pays for conference attendance, or the proportion of past employees that were poached from the firm via referrals. However, given the variables I have from the JSS, I choose to identify the type of a job based on the channel through which the worker received the job offer.

3.3 Simulated Method of Moments (SMM)

Since the JSS asks workers about offers they receive, and through which channel they received them, we observe the arrival rate of offers, and fix those rates from the data. So λ_R^U, λ_O^U, $\lambda_R^E(n = 0), \lambda_R^E(n = 1), \lambda_O^E$ are all fixed. Thus, we are left having to estimate the following parameters,

$$\theta = (\xi_{n=0}, \eta_{n=0}, \xi_{n=1}, \eta_{n=1}, \pi_n, p, c, \delta, b) \tag{11}$$

Where ξ_n and η_n are the parameters for each of the type-specific match productivity beta distributions. I calibrate these parameters using ten moments from the data, represented by the vector, $m^{data} \in \mathbb{R}^{10}$. To do this, I begin with any candidate parameter vector, θ, and solve for the equilibrium value functions. Then, using the solved model, a simulated panel is generated to approximate the stationary distribution of employment states, job types, and wages. From this simulated panel, a vector of corresponding model-implied moments, $\widehat{m}(\theta) \in \mathbb{R}^{10}$ is computed. Let $\ell(\theta)$ denote the vector of moment residuals, where for each moment k,

$$\ell_k(\theta) = \widehat{m}_k(\theta) - m_k^{\text{data}}, \tag{12}$$

Using this residual vector, the parameter vector is estimated as

$$\hat{\theta} = \arg\min_{\theta \in \Theta};\ \ell(\theta)^\top W \ell(\theta) \tag{13}$$

Where Θ is the parameter space and W is a positive semi-definite weighting matrix. In W, the wage-related moments are down-weighted relative to the non-wage moments. For the wage moments, rather than targeting the percentile values in levels, the estimator matches tail probabilities relative to the wage percentiles in the data. This eases estimation as the wage distribution has a

long upper tail.

In Table 1, I report the ten empirical moments alongside their corresponding data-implied moments using $\hat{\theta}$ for the Less than College and College+ sub-samples. The arrival rates taken from the data are listed in gray. For both education groups, the estimated model is able to match the unemployment rate, fraction of workers in network jobs, the EU and the EE flows, and the median wages quite well; however, the estimation struggles to match the percentiles of the wage distribution for both groups. In Table 2 the full set of parameters for the calibrated model is reported for both education groups.

Table 1: Moments from Data and Model Using Calibrated Parameters

	College+		Less than College	
Moment	**Model**	**Data**	**Model**	**Data**
Unemployment rate	0.046	0.045	0.061	0.062
Fraction employed in network jobs	0.342	0.343	0.362	0.359
*Official arrivals, unemployed (λ_O^U)	0.196	0.196	0.233	0.233
*Referral arrivals, unemployed (λ_O^R)	0.120	0.120	0.161	0.161
*Official arrivals, employed (λ_O^E)	0.148	0.148	0.162	0.162
*Referral arrivals, non-network ($\lambda_R^E(0)$)	0.056	0.056	0.052	0.052
*Referral arrivals, network ($\lambda_R^E(1)$)	0.084	0.084	0.084	0.084
Job destruction (EU) rate	0.011	0.011	0.009	0.013
Job-to-job (EE) transition rate	0.024	0.023	0.016	0.015
Median non-network wage	31.94	32.21	19.88	19.83
Median network wage	32.91	32.45	20.99	21.27
Mass of non-network wages below data p25	0.177	(21.59)	0.242	(14.09)
Mass of non-network wages above data p75	0.096	(47.88)	0.176	(28.85)
Mass of network wages below data p25	0.182	(22.11)	0.255	(15.16)
Mass of network wages above data p75	0.103	(49.47)	0.195	(29.55)

Notes: *Contact rates in gray taken from data. Data column for p25 and p75 reports the percentile value used as a threshold for the model tail probabilities.

Table 2: Estimated Model Parameters by Education Group

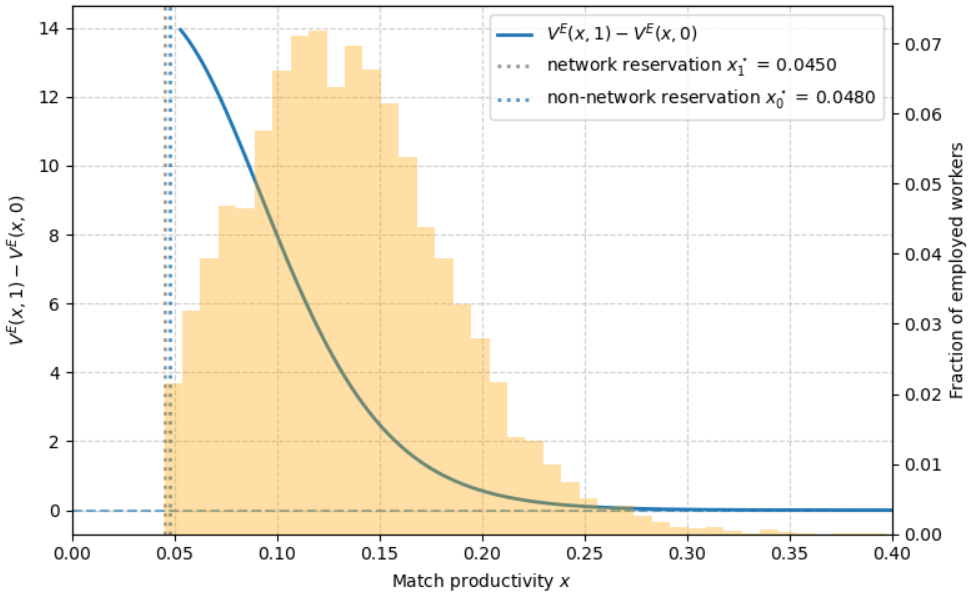
	College+	**Less than College**
b	5.58	1.95
p	403	352
c	-9.26	-12.9
δ	0.0082	0.0067
λ_O^U	0.196	0.233
λ_R^U	0.120	0.161
λ_O^E	0.148	0.162
$\lambda_R^E(n=0)$	0.056	0.052
$\lambda_R^E(n=1)$	0.084	0.084
$\xi_{n=0}$	1.08	0.852
$\eta_{n=0}$	26.3	27.4
$\xi_{n=1}$	0.979	0.853
$\eta_{n=1}$	23.5	24.7
π_n	0.323	0.308

4 Analysis of Calibrated Model

A key feature of the model is that the value of employment at a network job is state-dependent. For workers currently matched at low productivity, network jobs are especially valuable because they increase exposure to future job opportunities, raising the likelihood of upward job-to-job transitions. As a result, workers in low-productivity network jobs are willing to accept substantially lower wages than they would at comparable non-network jobs. In contrast, at higher levels of match productivity, the wage difference between network and non-network jobs is considerably smaller. This state dependence arises directly from the on-the-job search structure of the model. When a worker's current match productivity is low, the expected gain from receiving an additional offer is large, since most potential offers dominate the incumbent job. Network jobs raise the arrival rate of such offers, increasing the option value of remaining employed and accelerating movement up the job ladder. By contrast, workers already matched to high-productivity jobs face a smaller set of offers that strictly improve upon their current position, reducing the marginal benefit of additional search intensity. A real-world analogue is an apprenticeship or rotational position, in which workers accept lower initial pay in exchange for access to information and contacts that could lead to better jobs down the line.

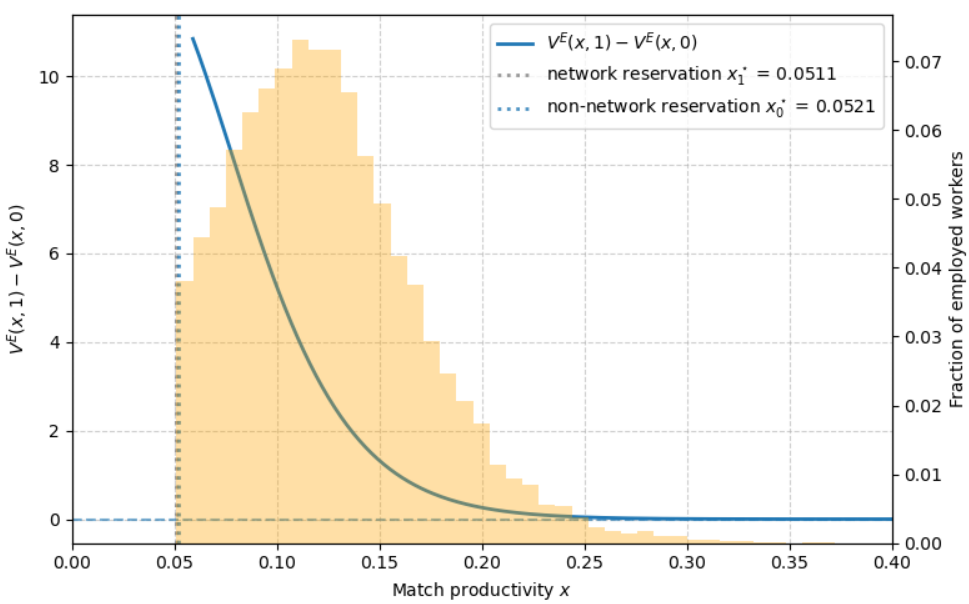
To illustrate this difference in how the network job is valued across the match-productivity distribution, Figure 1 and Figure 2 plot the difference in the employed value function between network and non-network jobs, $V^E(x, 1, f(x)) - V^E(x, 0, f(x))$, for the College+ and Less than College samples, respectively, under the simplifying assumption that wages equal marginal productivity ($w = f(x)$). The orange histogram in each figure shows the stationary distribution of employed workers over match productivity. As these figures illustrate, for workers at the lower end of the match productivity distribution, the network job premium is the largest. As a worker moves up

Figure 1: College+ : Value Function Premium for a Network Job



Notes: The blue solid line plots the value premium of network jobs, $V^E(x, 1, f(x)) - V^E(x, 0, f(x))$ (with $w = f(x)$), as a function of match productivity, x, for College+ workers in the estimated model. The vertical dotted line reports the reservation match productivity for network and non-network jobs, respectively. The orange histogram shows the stationary distribution of employed workers across match productivity levels on the right axis.

Figure 2: Less than College : Value Function Premium for a Network Job



Notes: The blue solid line plots the value premium of network jobs, $V^E(x, 1, f(x)) - V^E(x, 0, f(x))$ (with $w = f(x)$), as a function of match productivity, x, for less than college workers in the estimated model. The vertical dotted line reports the reservation match productivity for network and non-network jobs, respectively. The orange histogram shows the stationary distribution of employed workers across match productivity levels on the right axis.

the match productivity distribution, the additional value of the network job diminishes before completely flattening off at the upper tail of the match productivity distribution. This pattern is consistent with empirical evidence in Baranowska-Rataj et al. (2023), who show that low-wage workers benefit disproportionately from strong coworker networks, which facilitate transitions to better jobs through referrals and information transmission.

To quantify the role of network jobs and referrals in shaping equilibrium employment and wages, I conduct a series of counterfactual exercises in which specific arrival rates are shut down. Table 3 reports the benchmark employment rate and median wage of the estimated model for both the College+ and Less than College subsamples. To isolate the contribution of network jobs, I first re-solve the model setting $\pi_n = 0$, eliminating network jobs entirely. The resulting changes in employment and median wages are reported in the second rows of Table 3. In both education groups, the employment rate and median wages decline, but for the Less than College group, the size of the drops is nearly three times as large. As a second exercise, I resolve the model with the referrals completely removed from the model. With referrals shut down, the decrease in total employment and median wages relative to the no-network jobs counterfactual is much larger than the decrease from benchmark to the no-network jobs version of the model. Again, the decrease is larger for the Less than College group than it is for the College+ group; however, the magnitude of the difference between the two is not as large as in the case of just network jobs being removed. The greater importance of the network jobs and the referrals for the Less than College group supports the finding of Loury (2006) that workers with fewer outside options rely more on their networks and referrals.

Table 3: Contribution of Network Jobs and Referrals to Wages and Employment

Outcome	**Exercise**	**College+**	**Less than College**
Employment Rate	Benchmark	0.955	0.937
	No n Jobs	–0.24%	–0.64%
	No Referrals	–2.38%	–3.95%
Median Wages	Benchmark	\$32.83	\$20.18
	No n Jobs	–2.67%	–6.75%
	No Referrals	–14.20%	–20.12%

5 Concluding Remarks

The results of the calibrated model show the heterogeneous impact of network job and the referral channel by education group, as network jobs generate larger gains for Less than College workers, potentially by increasing exposure to job-to-job opportunities that would otherwise be difficult to access through formal search channels. For College+ workers, the increase in equilibrium wages and employment caused by network jobs and referrals is not as large, but the referral channel means workers climb the job ladder much faster. Both of these findings are in line with what has been found by Loury (2006) and Arbex et al. (2019). These asymmetric effects suggest that unequal ac-

cess to network-generating jobs may be an important channel through which education-level wage gaps persist. As greater access to network-generating jobs for lower education or lower skill workers has a much larger impact than for a higher education or higher skill worker Baranowska-Rataj et al. (2023).

It is important to acknowledge some key limitations of this model and its estimation. The model omits heterogeneity in worker ability, endogenous network formation, and firm-side heterogeneity, treating referral arrival rates as a fixed job type-dependent characteristic. While this choice is deliberate and allows the analysis to isolate job-provided networking opportunities, it rules out interaction between workers' past employment histories and their future access to referrals. In practice, network capital is likely to accumulate endogenously through repeated employment and would depend on job tenure.

A natural extension of this paper would be to consider a general equilibrium version of the model, where, rather than exogenously chosen firm types, firms providing networking opportunities become endogenous to the model, and the match-productivity distributions would depend on the arrival channel rather than job type. Meaning that referred workers would draw from a different distribution than workers who took the job through official or other channels. From the firm's perspective, being a network job would mean they hire more from this referral match-productivity distribution, but at the same time, workers already at this firm would exit more frequently, since they benefit from higher on-the-job arrival rates. Another approach I explored, but was unable to estimate using the JSS data, involved a sort of network stock that was bargained between the firm and the worker, along with the wage. To fully estimate this sort of model would require employee-firm panel data.

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